Given: Due:	Further Maths (Core Pure) HW 20	Name:

Chapter 4.5: Roots of polynomials (inc. transformations of roots)

You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.

91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)

Useful resources: Your notes from recent lessons & your text book

Questions - do these on lined paper

Q1. The cubic equation

$$2x^3 + 6x^2 - 3x + 12 = 0$$

has roots α, β and γ.

Without solving the equation, find the cubic equation whose roots are $(\alpha + 3)$, $(\beta + 3)$ and $(\gamma + 3)$, giving your answer in the form $pw^3 + qw^2 + rw + s = 0$, where p, q, r and s are integers to be found.

(5)

Q2.

The three solutions of the cubic equation

$$x^3 - 2x^2 + 3x + 1 = 0 \quad x \in \mathbb{R},$$

are denoted by α, β and γ.

Find a cubic equation with integer coefficients whose solutions are

$$2\alpha - 1, \; 2\beta - 1 \; \text{ and } \; 2\gamma - 1.$$

(4)

Q3.

The roots of the cubic equation

$$16x^3 - 8x^2 + 4x - 1 = 0 \quad x \in \mathbb{R},$$

are denoted in the usual notation by α, β and γ.

Find a cubic equation, with integer coefficients, whose roots are

$$\tfrac{4}{3}(\alpha - 1), \; \tfrac{4}{3}(\beta - 1) \; \text{ and } \; \tfrac{4}{3}(\gamma - 1).$$

(6)

Q4.

The polynomial $x^4 + x^3 - x + 12 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Find the polynomial with roots $\alpha^2, \beta^2, \gamma^2, \delta^2$. (4)

Q5.

The equation $x^4 - 6x^3 - 73x^2 + kx + m = 0$ has two positive roots, α, β and two negative roots γ, δ. It is given that $\alpha\beta = \gamma\delta = 4$.

(i) Find the values of the constants k and m. (4)

(ii) Show that $(\alpha + \beta)(\gamma + \delta) = -81$. (3)

(iii) Find the quadratic equation which has roots $\alpha + \beta$ and $\gamma + \delta$. (2)

(iv) Find $\alpha + \beta$ and $\gamma + \delta$. (2)

(v) Show that $\alpha^2 - 3(1+\sqrt{10})\alpha + 4 = 0$, and find similar quadratic equations satisfied by β, γ and δ. (3)

END OF HOMEWORK

Total = 33 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.